# Total Energy Conservation & Thermodynamic Applications

First Law of Thermodynamics for moving control volumes, enthalpy grouping, stagnation enthalpy $h_0$, and engineering shaft work.

OFFICIAL MATERIAL Notes.pdf · p. 35

## 1. The Master Total Energy Conservation Equation

The First Law of Thermodynamics for a closed material fluid system $V_f(t)$ states that the rate of increase of total energy (internal energy $e$ plus macroscopic kinetic energy $v^2/2$) equals the net mechanical power done on the fluid plus the net rate of heat transferred into the fluid:

$$\frac{d}{dt}\left[\int_{V_f(t)} \rho\left(e+\frac{|v|^2}{2}\right) dV\right] = \dot{W}_{\text{ext}} + \dot{Q}$$

*[Eq. 3.45] First Law of Thermodynamics for a closed material fluid system.*

Substituting intensive total energy $b = e + \frac{|v|^2}{2}$ into the universal Reynolds Transport Theorem yields the **Master Integral Total Energy Equation** for any moving control volume $V_c(t)$:

$$\mathbf{\frac{d}{dt}\left[\int_{V_c(t)} \rho\left(e+\frac{|\mathbf{v}|^2}{2}\right) d\mathbf{V}\right] + \int_{\Sigma_c(t)} \rho\left(e+\frac{|\mathbf{v}|^2}{2}\right)[(\mathbf{v}-\mathbf{v_c})\cdot\mathbf{n}]\, d\sigma = \dot{W}_{\text{ext}} + \dot{Q}}$$

*[Eq. 3.46] Master Integral Total Energy Conservation Equation.*

Here, the external power $\dot{W}_{\text{ext}}$ done on the fluid by boundary forces and body forces breaks down into:

$$\dot{W}_{\text{ext}} = -\int_{\Sigma_c} p(v\cdot n)\, d\sigma + \int_{\Sigma_c} v\cdot\bar{\bar{\tau}}'\cdot n\, d\sigma + \int_{V_c} \rho f_m \cdot v\, dV + \dot{W}_{\text{shaft}}$$

and total thermal heat transfer $\dot{Q}$ combines boundary conduction (Fourier's law $q = -k\nabla T$) and internal sources (chemical combustion $Q_c$ and radiative absorption $Q_r$):

$$\dot{Q} = -\int_{\Sigma_c} q \cdot n \, d\sigma + \int_{V_c} (Q_c + Q_r) \, dV$$

# Total Energy: Complete Derivation Chain (Notes.pdf, Eqs. 3.39-3.46)

## Step 1: Origin of each term of the First Law (Eq. 3.45)

The total energy of a fluid volume is $\int_{V_f} \rho(e + |v|^2/2) \, dV$. Its rate of change equals the power of the forces plus the heat added (First Law). Each contribution is built from an elementary power on a surface element $d\sigma$ or a volume element $dV$.

- **Surface forces.** The stress vector $(\bar{\bar{\tau}} \cdot n) \, d\sigma$ acts on material that moves with velocity $v$, so its power is $v \cdot \bar{\bar{\tau}} \cdot n \, d\sigma$. With $\bar{\bar{\tau}} = -p\bar{\bar{I}} + \bar{\bar{\tau}}'$ and $v \cdot (p\bar{\bar{I}}) \cdot n = p \, v \cdot n$, the total is $-\int p \, v \cdot n \, d\sigma + \int v \cdot \bar{\bar{\tau}}' \cdot n \, d\sigma$.
- **Body force.** The force $\rho f_m \, dV$ has power $\rho f_m \cdot v \, dV$.
- **Heat conduction.** By Eqs. 3.41 and 3.43, $q \cdot n \, d\sigma$ is the conductive heat leaving through $d\sigma$ (its divergence is the heat loss per unit volume), so the heat added is $-\int q \cdot n \, d\sigma$.
- **Volumetric sources.** Chemical reaction and radiation add $\int (Q_c + Q_r) \, dV$.

Summing these contributions gives Eq. 3.45:

$$\frac{d}{dt}\int_{V_f} \rho\Big(e + \frac{|v|^2}{2}\Big) dV = -\int_{\Sigma_f} p \, v \cdot n \, d\sigma + \int_{\Sigma_f} v \cdot \bar{\bar{\tau}}' \cdot n \, d\sigma + \int_{V_f} \rho f_m \cdot v \, dV - \int_{\Sigma_f} q \cdot n \, d\sigma + \int_{V_f} (Q_c + Q_r) \, dV$$

*Eq. 3.45*

## Step 2: Transfer to a moving control volume (Eq. 3.46)

Apply Eq. 3.8 with $\phi = \rho(e + |v|^2/2)$ to the left-hand side of Eq. 3.45 and evaluate the right-hand side on $\Sigma_c$ and $V_c$, which coincide with $\Sigma_f$ and $V_f$ at the instant considered. Moving the convective flux to the left gives

$$\frac{d}{dt}\int_{V_c} \rho\Big(e + \frac{|v|^2}{2}\Big) dV + \int_{\Sigma_c} \rho\Big(e + \frac{|v|^2}{2}\Big) \big[(v - v_c) \cdot n\big] d\sigma = -\int_{\Sigma_c} p \, v \cdot n \, d\sigma + \int_{\Sigma_c} v \cdot \bar{\bar{\tau}}' \cdot n \, d\sigma + \int_{V_c} \rho f_m \cdot v \, dV - \int_{\Sigma_c} q \cdot n \, d\sigma + \int_{V_c} (Q_c + Q_r) \, dV$$

*Eq. 3.46*

## Step 3: Flow work and specific enthalpy (extension)

**Extension, not in the Notes.** For a fixed control volume ($v_c = 0$), move the pressure work to the left-hand side and write $p \, v \cdot n = (p/\rho) \, \rho \, v \cdot n$:

$$\int_{\Sigma_0}\Big[\rho\Big(e+\frac{|v|^2}{2}\Big)+p\Big]v\cdot n\,d\sigma=\int_{\Sigma_0}\rho\Big(e+\frac{p}{\rho}+\frac{|v|^2}{2}\Big)v\cdot n\,d\sigma=\int_{\Sigma_0}\rho\Big(h+\frac{|v|^2}{2}\Big)v\cdot n\,d\sigma$$

with $h=e+p/\rho$. The balance for a fixed volume becomes

$$\int_{V_0}\frac{\partial}{\partial t}\Big[\rho\Big(e+\frac{|v|^2}{2}\Big)\Big]dV+\int_{\Sigma_0}\rho\Big(h+\frac{|v|^2}{2}\Big)v\cdot n\,d\sigma=\int_{\Sigma_0}v\cdot\bar{\bar{\tau}}'\cdot n\,d\sigma+\int_{V_0}\rho f_m\cdot v\,dV-\int_{\Sigma_0}q\cdot n\,d\sigma+\int_{V_0}(Q_c+Q_r)\,dV$$

For a moving boundary, $p\,v\cdot n=p\,(v-v_c)\cdot n+p\,v_c\cdot n$. The first part joins the enthalpy flux in the same way and the second part is the work of the pressure on the moving boundary, which stays on the right-hand side.

## Step 4: Gravity and the total enthalpy $h_0$ (extension)

**Extension, not in the Notes.** For gravity with $z$ pointing upward, $f_m=-\nabla(gz)$ (Eq. 3.14 with $U=gz$). In a steady flow, the continuity equation $\nabla\cdot(\rho v)=0$ (Eq. 4.8) lets the body-force power be written as a divergence:

$$\rho f_m\cdot v=-\rho v\cdot\nabla(gz)=-\nabla\cdot(\rho\,gz\,v)+gz\,\nabla\cdot(\rho v)=-\nabla\cdot(\rho\,gz\,v)$$

Gauss' theorem converts the volume integral into $-\int_{\Sigma_0}\rho\,gz\,v\cdot n\,d\sigma$. Moving it to the left-hand side adds $gz$ inside the convective flux:

$$\int_{\Sigma_0}\rho\,h_0\,v\cdot n\,d\sigma,\qquad h_0=h+\frac{|v|^2}{2}+gz$$

## Step 5: Steady engineering energy equation (extension)

**Extension, not in the Notes.** For a steady flow in a fixed control volume, split the boundary into ports, stationary solid walls and moving blades.

- On stationary walls the no-slip condition gives $v=0$, so the pressure and viscous work vanish; the heat conducted through the walls defines $\dot{Q}=-\int_{\text{walls}}q\cdot n\,d\sigma+\int(Q_c+Q_r)\,dV$.
- On the moving blades of a machine, the (time-averaged) power of pressure and viscous forces on the fluid is defined as $\dot{W}_{\text{shaft}}$.
- At the ports, viscous work and conduction are negligible against convection (high Reynolds and Peclet numbers) and $h_0$, $\rho$, $v$ are taken uniform, so $\int_{A_k}\rho h_0\,v\cdot n\,d\sigma=\dot{m}_k h_{0,k}$.

The balance of Step 4 then gives the form used in the engineering applications:

$$\sum_{\text{out}} |\dot{m}|\, h_0 - \sum_{\text{in}} |\dot{m}|\, h_0 = \dot{Q} + \dot{W}_{\text{shaft}}$$

Dimensional check: each term has units $\mathrm{kg\,s^{-1}} \cdot \mathrm{J\,kg^{-1}} = \mathrm{W}$.

## 2. Flow Work, Enthalpy, and Stagnation Enthalpy ($h_0$)

A crucial step in fluid thermodynamics is recognizing that the normal pressure work term $-\int_{\Sigma_c} p(v\cdot n)d\sigma$ can be written as:

$$-\int_{\Sigma_c} p(v\cdot n)\, d\sigma = -\int_{\Sigma_c} \frac{p}{\rho}\,\rho(v\cdot n)\, d\sigma$$

Moving this "flow work" (the work done by pressure pushing fluid packets across the control surface) to the left-hand side of the energy balance groups it seamlessly with internal energy $e$ to define **specific enthalpy** $h$ :

$$h = e + \frac{p}{\rho}$$

Combining specific enthalpy with kinetic energy and gravitational potential energy $gz$ defines the **Stagnation (Total) Enthalpy** $h_0$:

$$\mathbf{h_0 = h + \frac{|v|^2}{2} + gz}$$

For a calorically perfect gas ($h = c_pT$), the stagnation enthalpy is directly proportional to the **stagnation temperature** $T_0$ (here the $gz$ term of $h_0$ is neglected, which is customary for gases because $g\,\Delta z \ll c_pT$):

$$h_0 = c_pT_0 = c_pT + \frac{|v|^2}{2}$$

### The Engineering Energy Equation for 1D Streams

For steady flow through devices with discrete inlets and outlets, negligible viscous work at open ports, and steady shaft work $\dot{W}_{\text{shaft}}$ (from pumps, compressors, or turbines):

$$\dot{\mathbf{Q}} + \dot{\mathbf{W}}_{\text{shaft}} = \sum_{\mathbf{j}} \dot{\mathbf{m}}_{\text{out},\mathbf{j}} \mathbf{h}_{\mathbf{0},\text{out},\mathbf{j}} - \sum_{\mathbf{k}} \dot{\mathbf{m}}_{\text{in},\mathbf{k}} \mathbf{h}_{\mathbf{0},\text{in},\mathbf{k}}$$

SPECIAL CASES IN AEROSPACE PROPULSION

- **Adiabatic Duct ($\dot{Q} = 0, \dot{W}_{\text{shaft}} = 0$):** Stagnation enthalpy is strictly conserved: $h_{0,\text{out}} = h_{0,\text{in}}$. Kinetic energy increases only at the expense of static temperature ($T_{\text{out}} < T_{\text{in}}$).
- **Combustor ($\dot{W}_{\text{shaft}} = 0$):** Combustion heat release raises stagnation enthalpy: $\Delta h_0 = q_{\text{combustion}}$.
- **Compressor ($\dot{W}_{\text{shaft}} > 0$):** Mechanical work input raises total pressure and stagnation temperature: $w_{\text{in}} = c_p(T_{0,\text{out}} - T_{0,\text{in}})$.
- **Turbine ($\dot{W}_{\text{shaft}} < 0$):** Fluid delivers shaft power to drive the engine fan/compressor: $w_{\text{out}} = c_p(T_{0,\text{in}} - T_{0,\text{out}})$.

CANONICAL WALKTHROUGH 1 · CL15 SETUP

## Mixing Elbow Stream Exit Temperature $T_s$

**Physical Setup:** Two streams of an incompressible liquid (density $\rho$, specific heat $c_p$) enter an adiabatic, rigid mixing tee or elbow ($\dot{Q} = 0, \dot{W}_{\text{shaft}} = 0$). Stream 1 enters with mass flow rate $\dot{m}_1$ at temperature $T_1$ and speed $v_1$. Stream 2 enters with mass flow rate $\dot{m}_2$ at temperature $T_2$ and speed $v_2$. The mixed fluid exhausts into the atmosphere at state $s$ through an exit nozzle of area $A_s$.

**Objective:** Determine the exit mixed temperature $T_s$.

### 1 Mass Balance

From steady continuity:

$$\dot{m}_s = \dot{m}_1 + \dot{m}_2$$

The exit discharge velocity is $v_s = \frac{\dot{m}_s}{\rho A_s}$.

### 2 Total Energy Balance

For an adiabatic rigid mixer without shaft work ($\dot{Q} = 0, \dot{W}_{\text{shaft}} = 0$):

$$\sum \dot{m}_{\text{in}} h_{0,\text{in}} = \dot{m}_s h_{0,s} \implies \dot{m}_1 h_{0,1} + \dot{m}_2 h_{0,2} = (\dot{m}_1 + \dot{m}_2) h_{0,s}$$

Expanding the stagnation enthalpies with $h = cT$ ($c = c_p$ for an ideal gas; $c$ is the specific heat for a liquid):

$$\dot{m}_1 \left( c_p T_1 + \frac{v_1^2}{2} \right) + \dot{m}_2 \left( c_p T_2 + \frac{v_2^2}{2} \right) = (\dot{m}_1 + \dot{m}_2) \left( c_p T_s + \frac{v_s^2}{2} \right)$$

### 3 Solve for Exit Temperature $T_s$

**Assumption (note):** writing $h_0 = cT + v^2/2$ drops the pressure term, because for a liquid $h = e + p/\rho = cT + p/\rho$. The expression is valid for an ideal gas, where $h = c_p T$ exactly, or for a liquid only if the inlet and outlet pressures are about equal, so that the $p/\rho$ terms cancel in the balance. In low-speed liquid flow, kinetic energy differences are negligible compared to thermal energy ($v^2/2 \ll c_p T$). The energy equation reduces to the weighted calorimetric average:

$$\mathbf{T_s = \frac{\dot{m}_1 T_1 + \dot{m}_2 T_2}{\dot{m}_1 + \dot{m}_2}}$$

When discharge kinetic energy is non-negligible (high-speed nozzle discharge; for a liquid this formula is consistent only if the pressure differences are negligible compared with $hov^2$, since the neglected $p/ho$ terms are of the same order as $v^2/2$; for an ideal gas it is exact):

$$\mathbf{T_s = \frac{\dot{m}_1 T_1 + \dot{m}_2 T_2}{\dot{m}_1 + \dot{m}_2} + \frac{\dot{m}_1 v_1^2 + \dot{m}_2 v_2^2 - (\dot{m}_1 + \dot{m}_2) v_s^2}{2 c_p (\dot{m}_1 + \dot{m}_2)}}$$

CANONICAL WALKTHROUGH 2 · CL10 SETUP

# Ventilation Fan Nozzle Shaft Power $\dot{W}$

**Physical Setup:** An electric ventilation fan inside an adiabatic duct takes ambient air ($\rho, p_a, T_a$) from rest ($v_1 \approx 0$) and accelerates it through a contracting nozzle discharging into the room at high exit speed $v_e$, area $A_e$, at ambient pressure $p_e = p_a$. Total pressure loss due to friction across duct screens is $\Delta p_{\text{loss}}$.

**Objective:** Calculate the minimum mechanical shaft power $\dot{W}_{\text{shaft}}$ that must be supplied by the electric motor.

Analysis

## Energy Balance from Inlet to Exit

Mass flow rate through the nozzle: $\dot{m} = \rho A_e v_e$.

Applying the engineering energy equation across the fan and nozzle ($\dot{Q} = 0$, $z_1 \approx z_e$):

$$\dot{W}_{\text{shaft}} = \dot{m}(h_{0,e} - h_{0,1}) = \dot{m}\left[\left(h_e + \frac{v_e^2}{2}\right) - (h_1 + 0)\right]$$

For an incompressible air flow ($\rho = \text{const}$), enthalpy changes arise from the internal energy generated by viscous dissipation and from pressure differences ($p_e = p_1 = p_a \implies \Delta(p/\rho) = 0$). The loss is *defined* by the dissipated energy per unit mass, $\Delta p_{\text{loss}} \equiv \rho\,(e_e - e_1)$ (adiabatic duct, so all the dissipation stays in the fluid); this is a definition, not a derived result:

$$h_e - h_1 = (e_e - e_1) + \frac{p_e - p_1}{\rho} = \frac{\Delta p_{\text{loss}}}{\rho}$$

Substituting into the shaft power expression:

$$\dot{W}_{\text{shaft}} = \dot{m}\left(\frac{v_e^2}{2} + \frac{\Delta p_{\text{loss}}}{\rho}\right) = \mathbf{Q}\left(\frac{\mathbf{1}}{\mathbf{2}}\rho\mathbf{v_e^2} + \mathbf{\Delta p}_{\text{loss}}\right)$$

where $Q = A_e v_e$ is the volumetric airflow rate ($\text{m}^3/\text{s}$).

PHYSICAL ENERGY BREAKDOWN

The fan shaft power is partitioned into two clear physical duties:

1. **Kinetic energy flux ($\frac{1}{2}\rho Q v_e^2$):** The rate of kinetic energy carried away by the high-speed discharge jet.

2. **Viscous dissipation power ($Q\Delta p_{\text{loss}}$):** The mechanical work converted irrecoverably into microscopic internal thermal energy due to friction across fan blades and duct screens.

← 3.4 Angular Momentum

Topic 3 Hub

Next: 3.6 Stress Tensor & Friction →

Study notes written from the official course material (Notes.pdf · p. 35). Online version: subjects/fluid-mechanics/teoria/topic-3-5-energy.html